CTMT Morphism State on the Synthetic Coil Battery:
Inversion-Forced Recovery Class, Added Null/Composition Stress, and Honest Current Status

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Abstract

We report the current morphism state of CTMT on a synthetic coil battery under an explicitly conservative standard. Rather than assuming the desired admissible morphism class, we attempt to derive it from survival of inversion, Fisher geometry, resolved-subspace stability, and rupture stability. A first battery (V1) defined a recovery-forced class by calibration on gauge scenarios under local row-stochastic blur. A second battery (V2) added null-projector defect and composition-class variance in order to test whether the broader recovery class collapses to a narrower Čencov/transport-style class.

The result is mixed and therefore informative. Scenario-level separation remains non-trivial: baseline and current scaling occupy a high-survival regime, whereas radius scaling, compensated attack, causal violation, and unit leak occupy distinctly lower survival regimes. However, the added V2 stresses do *not* shrink the surviving morphism family beyond V1 on the present suite. Several non-Markov-looking families still survive cleanly on gauge scenarios while refusing non-gauge scenarios. Thus the current coil evidence supports only a broader inversion-forced *recovery-admissible* class, not the uniquely desired CTMT transport class. This paper states that conclusion explicitly, together with non-claims, falsification paths, and limitations.

Contents

1 Introduction and scope

This note documents the present status of CTMT morphism forcing on the synthetic coil suite. The goal is deliberately narrower than earlier transport notes. We do *not* claim to have derived the unique physically admissible morphism class. Instead, we ask a tested and falsifiable question:

> Can the coil inversion pipeline itself force a nontrivial admissible morphism class, and if so, how narrow is that class at the current stage of CTMT?

The answer is: yes, but only partially. The inversion battery defines a nontrivial class that preserves recovery on gauge scenarios and refuses several structural/violation scenarios. Yet, on the present suite, that class remains broader than the intended CTMT transport class.

2 Non-claims and conservative stance

This paper makes the following non-claims explicit.

(N1) We do not claim universality beyond the present synthetic coil family.

(N2) We do not claim that inversion forcing alone has derived the unique Čencov/Markov admissible class.

(N3) We do not claim that the present battery resolves the physical meaning of all surviving non-Markov morphism families.

(N4) We do not claim laboratory evidence. The battery is synthetic only.

3 Synthetic coil battery and inversion pipeline

The underlying forward model is the same local magnetostatic coil pipeline used in the recent CTMT transport suite: a circular coil with radius $R = 5\,\mathrm{cm}$, $n_z = 60$, $n_r = 40$, observation noise $\sigma_B = 2 \times 10^{-8}\,\mathrm{T}$, and anchor parameter

$$\theta_{\mathrm{anchor}} = (4.0,\ 5.8 \times 10^{-4},\ -4.2 \times 10^{-4}).$$

The local Jacobian J is obtained by finite differences of the B_z field, and parameter recovery is performed by local generalized least squares around the anchor.

The scenario suite is:

(S1) **baseline**;

(S2) **current scaling**;

(S3) **radius scaling**;

(S4) **compensated attack**;

(S5) **causal violation**;

(S6) **unit leak**.

The first two scenarios are gauge-candidate controls; the latter four are intended stress/violation constructions.

4 V1 and V2 forcing rules

4.1 V1: recovery-forced class

V1 calibrated an admissibility envelope on gauge scenarios (baseline and current scaling) under local row-stochastic blur. A candidate morphism survived only if all V1 defects stayed within the empirical 95% gauge envelope:

- $D_{\mathrm{inv}}^{\mathrm{mean}} \le 0.0351536$,
- $D_{\mathrm{inv}}^{\mathrm{max}} \le 0.0978566$,
- $D_F \le 0.00347277$,
- $D_{\mathrm{sub}} \le 0.487636$,
- $D_R \le 0.0357293$,
- $\kappa_{\mathrm{mid}} \le 7.39821 \times 10^7$.

4.2 V2: added null and composition stress

V2 retained the same calibration procedure and added two extra defects:

(V2-1) **Null-projector defect** D_{null}, measuring drift of the measurement-space null projector under the transformed Jacobian.

(V2-2) **Composition-class variance** D_{comp}, measuring whether repeated same-family compositions branch into visibly different normalized Fisher classes.

The V2 admissibility envelope was:

- $D_{\mathrm{inv}}^{\mathrm{mean}} \le 0.0351536$,
- $D_{\mathrm{inv}}^{\mathrm{max}} \le 0.0978566$,
- $D_F \le 0.00347277$,
- $D_{\mathrm{sub}} \le 0.487636$,
- $D_R \le 0.0357293$,
- $D_{\mathrm{null}} \le 0.468536$,
- $D_{\mathrm{comp}} \le 0.00232528$,
- $\kappa_{\mathrm{mid}} \le 7.39821 \times 10^7$.

A morphism-family instance survives V2 only if *all* V2 defects remain inside this gauge-calibrated envelope.

Definition 1 (Current inversion-forced class)**.** For the present synthetic coil suite, define $\mathcal{M}_{\mathrm{rec}}^{(1)}$ as the V1 survival class and $\mathcal{M}_{\mathrm{rec}}^{(2)}$ as the V2 survival class. These are tested classes only.

5 Main numerical outcomes

5.1 Scenario-level state

The V2 scenario summary is shown in Table 1.

Scenario	Survive	$D_{\text{inv}}^{\text{mean}}$	D_F	D_{sub}	D_R	D_{null}	D_{comp}
current_scaling	0.752	0.015	0.001	0.193	0.008	0.180	$3.95{\times}10^{-4}$
baseline	0.733	0.013	0.001	0.162	0.013	0.157	$3.65{\times}10^{-4}$
unit_leak	0.343	0.795	0.001	0.167	0.006	0.160	$6.93{\times}10^{-4}$
causal_violation	0.238	4.210	0.001	0.166	$2.03{\times}10^{-4}$	0.160	$3.95{\times}10^{-4}$
compensated_attack	0.238	95.739	0.001	0.165	0.016	0.159	$3.18{\times}10^{-4}$
radius_scaling	0.238	92.101	0.001	0.172	0.036	0.164	$4.15{\times}10^{-4}$

Table 1: V2 scenario-level summary. Baseline and current scaling remain the high-survival scenarios; the four remaining scenarios occupy a substantially lower-survival regime.

Two points are immediate. First, the gauge controls remain the only scenarios with survival rates above 0.7: baseline 0.733 and current scaling 0.752. Second, the structural/violation scenarios remain much lower: radius scaling 0.238, compensated attack 0.238, causal violation 0.238, and unit leak 0.343.

5.2 Family-level state and exact comparison V1 vs V2

The family-level V1/V2 comparison is shown in Table 2.

Family	V1 survive	V2 survive	D_{null} (V2)	D_{comp} (V2)
identity	1.000	1.000	0	0
causal_break	0.583	0.583	0.204	0
unit_leak	0.500	0.500	0.079	$4.78{\times}10^{-5}$
nonlocal_mix	0.475	0.475	0.072	$1.74{\times}10^{-4}$
markov_blur	0.417	0.417	0.240	$9.01{\times}10^{-4}$
sign_flip	0.000	0.000	0.263	$1.13{\times}10^{-3}$

Table 2: Family-level comparison of V1 and V2. The survive rates are numerically unchanged on the present suite.

The exact comparison is simple but important:

- V1 and V2 family survival rates are identical on the present suite.
- V1 and V2 scenario survival rates are also identical on the present suite.

In other words, adding null-projector stress and composition-class variance *did not shrink the surviving class* on the current coil battery.

5.3 Clean gauge/non-gauge split by family strength

The clean split criterion used here is strict: gauge-candidate average survival at least 0.9, while dangerous-compensated and violation averages are each at most 0.1. Under that rule, the current maximum clean strengths are:

Family	Maximum clean strength
causal_break	0.45
identity	–
markov_blur	0.30
nonlocal_mix	0.30
sign_flip	–
unit_leak	0.45

Table 3: Maximum family strength preserving a clean gauge/non-gauge split under V2.

The main observation from Table 3 is that several families survive the clean split well beyond the identity map: markov blur up to strength 0.30, nonlocal mixing up to 0.30, unit leak up to 0.45, and the present causal-break construction up to 0.45. Sign flips are refused at all sampled strengths.

6 Empirical interpretation

Remark 1 (Current coil-derived morphism state)**.** On the present synthetic coil battery, the inversion pipeline enforces a nontrivial recovery-admissible class that separates gauge scenarios from several structurally distinct violation scenarios. However, this class is not uniquely identified with the intended CTMT transport class, because several non-Markov morphism families remain admissible under the calibrated constraints.

Interpretation. This statement is based on numerical evidence only and does not constitute a formal theorem. It summarizes the observed stability and failure patterns of the inversion pipeline under the tested morphism families.

Corollary 1 (Honest current conclusion)**.** *The strongest conclusion currently supported is that CTMT has derived a tested* recovery-admissible *class on the coil suite, not yet a uniquely justified transport/Čencov admissible class.*

Statistical caution. All reported survival rates are empirical averages over a finite number of runs (five samples per configuration in the present battery). No confidence intervals or bootstrap estimates are reported. Differences such as 0.752 vs. 0.733 should therefore not be interpreted as statistically significant without further analysis.

7 Choice of morphism families

The tested morphism families are not claimed to span the full space of admissible or inadmissible transformations. They represent distinct qualitative deformations:

- local stochastic blur (markov blur),
- nonlocal redistribution (nonlocal mixing),
- multiplicative scaling defects (unit leak),
- temporal inconsistency (causal break),
- sign-structure violation (sign flip).

No claim of completeness is made. Additional untested morphism families may exist that alter the present conclusions.

8 What failed to improve under V2

The intended role of V2 was to make the battery narrower by adding null-projector and composition consistency stress. On the present suite, that intended narrowing *failed.* The additional V2 metrics remained too mild relative to the nuisance families that already survived V1. This failure is substantive and should be reported, not hidden, because it marks the actual boundary of the present derivation attempt.

In particular:

(F1) V2 did not eliminate nonlocal mixing from the clean gauge/non-gauge split.

(F2) V2 did not eliminate the present unit-leak family from the clean gauge/non-gauge split.

(F3) V2 did not eliminate the present causal-break family from the clean gauge/non-gauge split.

(F4) Only sign-flip style corruption is uniformly refused already at the family level.

9 Failure of V2 sharpening

The addition of V2 constraints does not reduce the admissible class on the present battery.

- Family survival rates are unchanged.
- Scenario-level separation is unchanged.
- Surviving nuisance families remain admissible.

This indicates that the added constraints are not sufficiently discriminative in the current experimental setting.

This is a negative but informative result.

10 Falsifications and sharpening paths

The present state is falsifiable in both directions.

Proposition 1 (Downward falsification)**.** *If future batteries show that even gauge scenarios cannot maintain high survival under clearly admissible local row-stochastic coarse-graining, then the current recovery-forcing claim fails.*

Proposition 2 (Upward sharpening)**.** *If future batteries add stronger quotient collapse, path-composition, or null-transport tests and these systematically eliminate the nuisance surviving families while preserving gauge controls, then the present recovery-admissible class may be sharpened toward a narrower CTMT transport class.*

11 Limitations

(L1) The battery is synthetic and local around a fixed anchor.

(L2) The present causal-break family is a test construction, not a theorem-level characterization of physical causality.

(L3) Composition-class variance is implemented only through normalized Fisher class spread under repeated same-family compositions; it is therefore still a weak quotient proxy.

(L4) Null-projector defect is measured on the measurement-side Jacobian geometry only; it does not yet fully encode the stronger null-transport structure used in earlier CTMT transport notes.

(L5) The numerical thresholds are empirical 95% calibration envelopes, not analytic sharp constants.

12 Conclusion

The current morphism state of CTMT on the synthetic coil battery is now clearer than before. The inversion pipeline does force a nontrivial class: gauge scenarios survive at high rates, several structural/violation scenarios do not, and sign-flip corruption is uniformly refused. However, the stronger claim one would ultimately want — that inversion plus added structural stress uniquely determines the intended CTMT transport class — is not yet supported. The added V2 null/composition stress did not narrow the surviving family beyond V1 on this suite.

That is the honest current status. CTMT presently supports a coil-tested recovery-admissible class, not yet a uniquely derived transport/Čencov class. This is still useful: it identifies exactly what has been gained, exactly what remains missing, and exactly where the next falsifiable battery must attack.

References

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